

RNA-seq Production Workflow Student Pack

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RNA-seq Tool Selection Policy For Survom

Purpose

Survom is a production multiomics platform, not a tool encyclopedia. Each workflow step should expose one recommended default and, only when justified, one fallback or special-case tool.

The policy below keeps workflows understandable for guided users, predictable for production support, and reusable across multiomics assays.

Core Rule

Each step may define at most:

- **Default tool / algorithm**
- **Fallback tool / algorithm**

If one tool is enough, no fallback is required.

Selection Criteria

Prefer tools that have:

- stable, machine-readable outputs,
- MultiQC support where relevant,
- strong community adoption,
- simple parameter surface,
- active maintenance,
- mature Docker/Conda packaging,
- predictable resource use,
- clear Nextflow DSL2 module boundaries,
- strong compatibility with downstream steps.

Avoid exposing tools only because they are popular in the ecosystem. More options create more validation cases, more UI complexity, more documentation burden, and more support risk.

Supported RNA-seq MVP Routes

Route A: Alignment-Based Gene-Level Analysis

FASTQ -> QC -> trimming -> STAR -> samtools sort/index -> featureCounts -> DESeq2 -> enrichment

Use this when:

- BAM files are required,
- genome alignment QC is important,
- splice junction information matters,
- downstream visualization in IGV is expected,
- users need conventional gene-level counts from aligned reads.

Route B: Pseudoalignment Transcript/Gene-Level Analysis

FASTQ -> QC -> trimming -> Salmon -> tximport -> DESeq2 -> enrichment

Use this when:

- the primary goal is expression quantification,
- speed and lower storage are important,
- BAM output is not required,
- transcript-level estimates may be useful.

Default Survom route for standard bulk RNA-seq MVP:

Route B: Salmon -> tximport -> DESeq2

Reason: faster, easier to scale, less storage-heavy, and sufficient for many differential expression workflows.

Curated Tool Choices

Task	Default	Fallback	Rationale
Project initialization	custom validator	none	Platform-specific
Sample sheet validation	Python/Pydantic or JSON Schema	none	Keep schema enforcement centralized
FASTQ integrity	Nextflow staging + md5/gzip checks	seqkit stats	seqkit only when deeper FASTQ stats are needed
Raw FASTQ QC	FastQC	Falco	Falco only for speed/large-scale cases
Trimming	fastp	Cutadapt	fastp is simple; Cutadapt handles exact custom adapters
Post-trim QC	FastQC	Falco	Same as raw QC
Strandedness inference	Salmon auto lib type	RSeQC infer_experiment.py	RSeQC requires BAM, so fallback only
Reference validation	custom reference bundle validator	none	Platform-specific
Alignment	STAR	HISAT2	STAR default for bulk RNA-seq; HISAT2 for lower memory/special cases
BAM sort/index	samtools	none	Stable standard
Gene counting	featureCounts	none	Stable, widely used
Pseudoalignment	Salmon	Kallisto	Salmon default for richer output and mature workflow use
Transcript-to-gene summarization	tximport	tximeta	tximeta when provenance-aware transcriptome metadata is needed
Differential expression	DESeq2	edgeR	DESeq2 default; edgeR for specific designs/count models
ORA enrichment	clusterProfiler ORA	g:Profiler API/offline export	ORA is a distinct analytical task
GSEA enrichment	fgsea	clusterProfiler GSEA	GSEA is separate from ORA
PCA/sample QC	DESeq2/R base PCA	plotly export	Plotly only for interactive product view
Volcano plot	EnhancedVolcano	ggplot2 custom	EnhancedVolcano default for standard output
Heatmap	pheatmap	ComplexHeatmap	ComplexHeatmap for complex annotations
MultiQC aggregation	MultiQC	none	Production standard
Report rendering	Quarto/R Markdown	none	Reproducible reporting
Provenance capture	custom run manifest	RO-Crate export	RO-Crate for advanced interoperability

Parameter Exposure Policy

Use three parameter levels:

User-Settable Parameters

Visible in guided and dynamic UI.

Examples:

- organism,
- genome build,
- library layout,
- strandedness,
- route choice,
- condition column,
- batch column,
- FDR threshold.

Expert Parameters

Hidden under advanced settings.

Examples:

- STAR index overhang,
- Salmon library type override,
- featureCounts strand mode,
- DESeq2 design formula,
- minimum count filtering threshold.

Internal Parameters

Controlled by platform defaults and not normally shown.

Examples:

- Nextflow channel names,
- publish directory structure,
- container paths,
- retry policy,
- temporary work directory.

Verification Policy

If a command-line flag, default, or tool behavior is uncertain, write:

VERIFY_IN_OFFICIAL_DOCS

Do not invent exact flags or defaults.

Multomics Reuse Tags

Every step should be tagged as one of:

- COMMON
- RNA_SEQ_SPECIFIC

- GENOMICS_COMMON
- STATISTICS_COMMON
- REPORTING_COMMON

This supports reusing common steps across RNA-seq, genomics, ATAC-seq, ChIP-seq, scRNA-seq, proteomics where appropriate, metabolomics where applicable, microbiome, and methylation workflows.

Curated RNA-seq Atomic Step Registry For Survom

This registry defines a clean production-oriented RNA-seq workflow model for Survom.

It intentionally avoids listing every possible tool. Each step has one default tool or algorithm and, only when useful, one fallback or special-case tool.

Supported MVP routes:

- **Route A:** FASTQ -> QC -> trimming -> STAR -> samtools -> featureCounts -> DESeq2 -> enrichment
- **Route B:** FASTQ -> QC -> trimming -> Salmon -> tximport -> DESeq2 -> enrichment

Both routes require sample-sheet validation and reference-bundle validation before quantification or alignment.

Default route for standard bulk RNA-seq:

Route B: Salmon -> tximport -> DESeq2

00 Project Initialization

```

step_id: rnaseq_00_project_initialization
step_name: Project initialization
category: COMMON
purpose: Create a reproducible run context before any data processing starts.
why_this_step_exists: Production runs need a stable run ID, output directory, execution
  profile, provenance record, and parameter snapshot.
required_inputs:
  - project_name
  - organism
  - genome_build
  - outdir
optional_inputs:
  - profile
  - resume
  - container_engine
input_formats:
  - UI form
  - JSON
  - YAML
required_outputs:
  - run_manifest
  - project_run_directory
  - resolved_params
optional_outputs:
  - audit_event
output_formats:
  - JSON
  - directory

```

```

depends_on: []
can_be_skipped: false
skip_conditions: []
default_tool_or_algorithm: custom project config validator
fallback_tool_or_algorithm: none
tool_selection_logic: Always use the platform validator.
user_settable_parameters:
  - project_name
  - organism
  - genome_build
  - outdir
  - profile
  - resume
  - container_engine
  - max_cpus
  - max_memory
  - max_time
expert_parameters:
  - work_dir
  - publish_mode
internal_parameters:
  - run_id
  - audit_actor
  - registry_version
qc_metrics:
  - config_valid
qc_pass_fail_rules:
  - fail if required project metadata is missing
  - fail if output directory is not writable
common_failure_modes:
  - invalid organism/genome build pair
  - unwritable output directory
  - unsupported execution profile
nextflow_module_name: platform/init_project
reusable_for_other_omics: COMMON
downstream_steps:
  - rnaseq_01_sample_sheet_validation
notes_for_guided_flowchart: Show this as project setup, not as a scientific analysis step.
notes_for_dynamic_flowchart: Lock this as the root node.
notes_for_chat_mode: Chat may collect missing project metadata before drafting a workflow.

```

01 Sample Sheet Validation

```

step_id: rnaseq_01_sample_sheet_validation
step_name: Sample sheet validation
category: COMMON
purpose: Validate sample metadata and map biological samples to input files.
why_this_step_exists: Most RNA-seq failures come from incorrect sample metadata, missing
  files, duplicated sample IDs, or invalid condition labels.
required_inputs:
  - sample_sheet
optional_inputs:
  - project_metadata
input_formats:
  - CSV
  - TSV
required_outputs:
  - validated_sample_sheet

```

```
- sample_manifest
optional_outputs:
- validation_report
output_formats:
- CSV
- JSON
- HTML
depends_on:
- rnaseq_00_project_initialization
can_be_skipped: false
skip_conditions: []
default_tool_or_algorithm: Python/Pydantic or JSON Schema validator
fallback_tool_or_algorithm: none
tool_selection_logic: Use one platform schema implementation to keep validation consistent
  across UI, API, and Nextflow.
user_settable_parameters:
- sample_sheet
- sample_id
- fastq_1
- fastq_2
- condition
- batch
- replicate
- strandedness
- library_layout
expert_parameters:
- allow_single_replicate_warning
- allow_missing_batch_column
internal_parameters:
- schema_version
- file_resolver
qc_metrics:
- number_of_samples
- number_of_conditions
- replicates_per_condition
- missing_fastq_count
- duplicate_sample_id_count
qc_pass_fail_rules:
- fail if FASTQ paths do not exist
- fail if sample IDs are not unique
- fail if condition column is missing
- fail if paired-end files are not correctly paired
- warn if fewer than 3 biological replicates per condition
- warn if batch column is missing
- warn or fail if batch is confounded with condition
common_failure_modes:
- sample IDs contain spaces or unsafe characters
- condition labels inconsistent by capitalization
- R1/R2 naming mismatch
- metadata says paired-end but only one FASTQ is present
nextflow_module_name: common/validate_samplesheet
reusable_for_other_omics: COMMON
downstream_steps:
- rnaseq_02_fastq_staging_integrity
notes_for_guided_flowchart: Show actionable errors in plain language.
notes_for_dynamic_flowchart: This node must produce a typed sample_manifest before FASTQ
  steps unlock.
notes_for_chat_mode: Chat can ask for missing condition, batch, or replicate columns.
```

02 FASTQ Staging And Integrity

```
step_id: rnaseq_02_fastq_staging_integrity
step_name: FASTQ staging and integrity
category: COMMON
purpose: Stage FASTQ files into the workflow and confirm they are readable before compute-
  heavy steps.
why_this_step_exists: Broken gzip files, incomplete uploads, or path resolution errors
  should fail early.
required_inputs:
  - sample_manifest
  - raw_fastq_files
optional_inputs:
  - md5_manifest
input_formats:
  - FASTQ.GZ
  - FASTQ
  - MD5 text
required_outputs:
  - staged_fastq_manifest
  - integrity_report
optional_outputs:
  - fastq_stats
output_formats:
  - JSON
  - TSV
depends_on:
  - rnaseq_01_sample_sheet_validation
can_be_skipped: false
skip_conditions: []
default_tool_or_algorithm: Nextflow channel staging plus md5/gzip integrity check
fallback_tool_or_algorithm: seqkit stats
tool_selection_logic: Use built-in staging and lightweight integrity checks by default; use
  seqkit when deeper read statistics are needed.
user_settable_parameters:
  - verify_md5
expert_parameters:
  - run_seqkit_stats
internal_parameters:
  - staged_file_paths
  - storage_backend
qc_metrics:
  - files_staged
  - gzip_integrity_failures
  - md5_mismatches
  - read_count_estimate
qc_pass_fail_rules:
  - fail if any required FASTQ cannot be staged
  - fail if gzip integrity check fails
  - fail if provided md5 does not match
common_failure_modes:
  - interrupted upload
  - incorrect object storage path
  - uncompressed FASTQ too large for configured policy
nextflow_module_name: common/stage_fastq_integrity
reusable_for_other_omics: COMMON
downstream_steps:
  - rnaseq_03_raw_read_qc
notes_for_guided_flowchart: Hide implementation details; show "files verified".
```

notes_for_dynamic_flowchart: Emits typed fastq_pairs or fastq_single.
notes_for_chat_mode: Chat should explain missing or corrupt files before workflow generation.

03 Raw Read Quality Control

```
step_id: rnaseq_03_raw_read_qc
step_name: Raw read quality control
category: COMMON
purpose: Assess sequencing quality before modifying reads.
why_this_step_exists: Raw QC identifies adapter contamination, low-quality cycles, GC bias,
  duplication, and overrepresented sequences.
required_inputs:
  - staged_fastq_manifest
optional_inputs:
  - custom_adapters
input_formats:
  - FASTQ.GZ
  - FASTQ
required_outputs:
  - fastqc_html
  - fastqc_zip
optional_outputs:
  - raw_qc_summary
output_formats:
  - HTML
  - ZIP
  - JSON
depends_on:
  - rnaseq_02_fastq_staging_integrity
can_be_skipped: false
skip_conditions: []
default_tool_or_algorithm: FastQC
fallback_tool_or_algorithm: Falco
tool_selection_logic: Use FastQC by default for mature MultiQC compatibility; use Falco for
  large cohorts where FastQC runtime is the bottleneck.
user_settable_parameters:
  - threads
expert_parameters:
  - custom_adapters
  - custom_contaminants
internal_parameters:
  - output_prefix
qc_metrics:
  - per_base_quality_status
  - adapter_content_status
  - gc_content_status
  - duplication_status
  - overrepresented_sequences_status
qc_pass_fail_rules:
  - warn if adapter content is high
  - warn if per-base quality degrades strongly at read ends
  - warn if GC distribution is abnormal
  - do not fail solely on FastQC warnings without user review
common_failure_modes:
  - wrong file type
  - corrupted FASTQ
  - unexpected library contamination
```

```
nextflow_module_name: common/fastqc
reusable_for_other_omics: COMMON
downstream_steps:
  - rnaseq_04_adapter_quality_trimming
  - rnaseq_07_salmon_quantification
  - rnaseq_09_star_alignment
notes_for_guided_flowchart: Explain QC warnings but do not overload users with all FastQC
modules.
notes_for_dynamic_flowchart: This node can feed trimming or direct quantification if
trimming is skipped.
notes_for_chat_mode: Chat should summarize QC issues and recommend whether trimming is
needed.
```

04 Adapter And Quality Trimming

```
step_id: rnaseq_04_adapter_quality_trimming
step_name: Adapter and quality trimming
category: COMMON
purpose: Remove adapters and low-quality bases that can reduce quantification or alignment
quality.
why_this_step_exists: Adapter contamination and poor-quality read tails can bias downstream
mapping and counting.
required_inputs:
  - staged_fastq_manifest
optional_inputs:
  - custom_adapter_sequences
input_formats:
  - FASTQ.GZ
  - FASTQ
required_outputs:
  - trimmed_fastq_manifest
  - trimming_report
optional_outputs:
  - trimming_html
  - trimming_json
output_formats:
  - FASTQ.GZ
  - HTML
  - JSON
depends_on:
  - rnaseq_03_raw_read_qc
can_be_skipped: true
skip_conditions:
  - raw QC shows no adapter contamination and high read quality
  - user selects no trimming with expert approval
default_tool_or_algorithm: fastp
fallback_tool_or_algorithm: Cutadapt
tool_selection_logic: fastp is default because it combines adapter detection, quality
filtering, and machine-readable reports. Cutadapt is fallback for exact custom adapters or
reproducibility with older studies.
user_settable_parameters:
  - minimum_read_length
  - quality_threshold
  - trim_poly_g
  - trim_poly_x
expert_parameters:
  - explicit_adapter_r1
  - explicit_adapter_r2
```

```
- front_trim_r1
- front_trim_r2
internal_parameters:
- threads
- report_prefix
qc_metrics:
- reads_before_filtering
- reads_after_filtering
- percent_reads_retained
- bases_trimmed
- adapter_trimmed_reads
qc_pass_fail_rules:
- warn if too many reads are discarded
- warn if adapter content remains high after trimming
- fail if output FASTQ is empty
common_failure_modes:
- over-trimming due to incorrect adapter sequence
- minimum read length too strict
- mixed library types in one sample sheet
nextflow_module_name: common/fastp
reusable_for_other_omics: COMMON
downstream_steps:
- rnaseq_05_post_trim_qc
notes_for_guided_flowchart: Default to fastp and expose only simple quality/min-length
controls.
notes_for_dynamic_flowchart: Trimming may be skipped if QC and route allow it.
notes_for_chat_mode: Chat should recommend trimming only when QC or protocol supports it.
```

05 Post-Trim Quality Control

```
step_id: rnaseq_05_post_trim_qc
step_name: Post-trim quality control
category: COMMON
purpose: Confirm trimming improved reads and generated usable FASTQ files.
why_this_step_exists: Users need pre/post trimming comparison before alignment or
quantification.
required_inputs:
- trimmed_fastq_manifest
optional_inputs: []
input_formats:
- FASTQ.GZ
required_outputs:
- post_trim_fastqc_html
- post_trim_fastqc_zip
optional_outputs:
- post_trim_qc_summary
output_formats:
- HTML
- ZIP
- JSON
depends_on:
- rnaseq_04_adapter_quality_trimming
can_be_skipped: true
skip_conditions:
- trimming was skipped
default_tool_or_algorithm: FastQC
fallback_tool_or_algorithm: Falco
tool_selection_logic: Same as raw read QC.
```

```

user_settable_parameters:
  - threads
expert_parameters: []
internal_parameters:
  - output_prefix
qc_metrics:
  - adapter_content_status_after_trimming
  - per_base_quality_status_after_trimming
qc_pass_fail_rules:
  - warn if adapter content remains high
  - warn if read quality remains poor
common_failure_modes:
  - trimming did not remove adapter due to unusual adapter sequence
  - trimmed reads too short
nextflow_module_name: common/fastqc_post_trim
reusable_for_other_omics: COMMON
downstream_steps:
  - rnaseq_06_strandedness_inference
  - rnaseq_07_salmon_quantification
  - rnaseq_09_star_alignment
notes_for_guided_flowchart: Show pre/post QC comparison.
notes_for_dynamic_flowchart: Output is QC-only; it does not change reads.
notes_for_chat_mode: Chat should explain whether trimming improved the data.

```

06 Strandedness Inference

```

step_id: rnaseq_06_strandedness_inference
step_name: Strandedness inference
category: RNA_SEQ_SPECIFIC
purpose: Determine whether the library is unstranded, forward-stranded, or reverse-stranded.
why_this_step_exists: Incorrect strandedness causes systematically wrong quantification and
  can silently invalidate results.
required_inputs:
  - fastq_manifest
  - transcriptome_index
optional_inputs:
  - aligned_bam
  - bed12_annotation
input_formats:
  - FASTQ.GZ
  - Salmon index
  - BAM
  - BED12
required_outputs:
  - strandedness_call
  - strandedness_evidence
optional_outputs:
  - inference_log
output_formats:
  - JSON
  - TXT
depends_on:
  - rnaseq_05_post_trim_qc
can_be_skipped: true
skip_conditions:
  - strandedness is known and explicitly supplied in sample sheet
default_tool_or_algorithm: Salmon auto library type on representative subsample
fallback_tool_or_algorithm: RSeQC infer_experiment.py

```

```
tool_selection_logic: Salmon is default because it can infer library type without a BAM.  
RSeQC is fallback when BAM exists or transcriptome indexing is unavailable.  
user_settable_parameters:  
- strandedness  
expert_parameters:  
- subsample_read_count  
- representative_samples_per_batch  
internal_parameters:  
- inference_threshold  
qc_metrics:  
- compatible_fragment_ratio  
- strandedness_confidence  
qc_pass_fail_rules:  
- fail if auto-inference is ambiguous and strandedness is required  
- warn if batches disagree in inferred strandedness  
common_failure_modes:  
- wrong transcriptome index  
- low-complexity subsample  
- mixed library prep protocols  
nextflow_module_name: rnaseq/infer_strandedness  
reusable_for_other_omics: RNA_SEQ_SPECIFIC  
downstream_steps:  
- rnaseq_07_salmon_quantification  
- rnaseq_11_featurecounts_gene_counting  
notes_for_guided_flowchart: Present as "library orientation check".  
notes_for_dynamic_flowchart: Required when strandedness is set to auto.  
notes_for_chat_mode: Chat should ask for lab protocol if inference is ambiguous.
```

06A Reference Bundle Validation

```
step_id: rnaseq_06a_reference_bundle_validation  
step_name: Reference bundle validation  
category: GENOMICS_COMMON  
purpose: Confirm that genome, transcriptome, annotation, and prebuilt indexes are mutually  
compatible before alignment or quantification.  
why_this_step_exists: Wrong genome builds, mismatched GTF files, transcript ID version  
mismatches, or stale indexes can silently corrupt RNA-seq results.  
required_inputs:  
- organism  
- genome_build  
- selected_route  
optional_inputs:  
- genome_fasta  
- gtf_annotation  
- star_index  
- transcriptome_fasta  
- salmon_index  
- tx2gene_mapping  
input_formats:  
- FASTA  
- GTF  
- STAR index directory  
- Salmon index directory  
- CSV  
- TSV  
required_outputs:  
- validated_reference_bundle  
- reference_validation_report
```

```
optional_outputs:
  - index_build_recommendation
output_formats:
  - JSON
  - HTML
depends_on:
  - rnaseq_00_project_initialization
can_be_skipped: false
skip_conditions: []
default_tool_or_algorithm: custom Survom reference bundle validator
fallback_tool_or_algorithm: none
tool_selection_logic: Reference compatibility must be enforced by the platform because it
  spans files, indexes, and metadata rather than one external tool.
user_settable_parameters:
  - organism
  - genome_build
  - selected_route
expert_parameters:
  - allow_custom_reference
  - allow_index_reuse
internal_parameters:
  - reference_registry_version
  - expected_chromosome_style
  - transcript_id_version_policy
qc_metrics:
  - reference_files_present
  - chromosome_name_compatibility
  - transcript_gene_mapping_rate
  - index_matches_reference
qc_pass_fail_rules:
  - fail if selected route lacks its required index
  - fail if genome FASTA and GTF chromosome naming are incompatible
  - fail if Salmon index, transcriptome, and tx2gene mapping are incompatible
  - warn if custom reference lacks provenance
common_failure_modes:
  - hg19/hg38 mismatch
  - Ensembl/UCSC chromosome naming mismatch
  - transcript IDs with version suffixes do not match tx2gene table
  - STAR or Salmon index built from a different reference release
nextflow_module_name: common/validate_reference_bundle
reusable_for_other_omics: GENOMICS_COMMON
downstream_steps:
  - rnaseq_07_salmon_quantification
  - rnaseq_09_star_alignment
notes_for_guided_flowchart: Show this as "reference checked" and surface only actionable
  mismatches.
notes_for_dynamic_flowchart: Must emit a validated reference bundle before alignment or
  quantification nodes unlock.
notes_for_chat_mode: Chat should ask for organism/genome build when missing and warn about
  custom reference provenance.
```

07 Salmon Quantification

```
step_id: rnaseq_07_salmon_quantification
step_name: Transcript quantification with Salmon
category: RNA_SEQ_SPECIFIC
purpose: Quantify transcript abundance directly from FASTQ reads.
```

```
why_this_step_exists: Provides fast transcript-level quantification without generating large
BAM files.
required_inputs:
  - analysis_fastq_manifest
  - transcriptome_index
  - strandedness_call
optional_inputs:
  - gene_transcript_map
input_formats:
  - FASTQ.GZ
  - Salmon index
  - JSON
required_outputs:
  - salmon_quant_files
  - salmon_logs
optional_outputs:
  - mapping_metrics
output_formats:
  - quant.sf
  - JSON
  - LOG
depends_on:
  - rnaseq_03_raw_read_qc
  - rnaseq_06_strandedness_inference
  - rnaseq_06a_reference_bundle_validation
can_be_skipped: true
skip_conditions:
  - user chooses alignment-based route
default_tool_or_algorithm: Salmon
fallback_tool_or_algorithm: Kallisto
tool_selection_logic: Salmon is default for mature output, bias correction support, and
common tximport workflows. Kallisto is fallback for studies that require compatibility with
existing Kallisto outputs.
user_settable_parameters:
  - quant_route
  - strandedness
expert_parameters:
  - bias_correction
  - gc_bias_correction
  - validate_mappings
internal_parameters:
  - threads
  - index_path
qc_metrics:
  - mapping_rate
  - fragments_processed
  - library_type_compatibility
qc_pass_fail_rules:
  - warn if mapping rate is low
  - fail if no quantification files are produced
  - fail if library type is incompatible with inferred strandedness
common_failure_modes:
  - wrong transcriptome index
  - incorrect strandedness
  - poor read quality
nextflow_module_name: rnaseq/salmon_quant
reusable_for_other_omics: RNA_SEQ_SPECIFIC
downstream_steps:
  - rnaseq_08_tximport_gene_summarization
notes_for_guided_flowchart: Default route for standard expression analysis.
```

notes_for_dynamic_flowchart: Emits transcript_quant and can feed tximport.
notes_for_chat_mode: Chat should choose this route for fast differential expression unless BAM is requested.

08 Transcript-To-Gene Summarization

```
step_id: rnaseq_08_tximport_gene_summarization
step_name: Transcript-to-gene summarization
category: RNA_SEQ_SPECIFIC
purpose: Convert transcript-level quantification to gene-level counts suitable for
         differential expression.
why_this_step_exists: DESeq2 gene-level analysis needs a consistent gene-by-sample matrix.
required_inputs:
  - salmon_quant_files
  - tx2gene_mapping
optional_inputs:
  - transcriptome_metadata
input_formats:
  - quant.sf
  - CSV
  - TSV
required_outputs:
  - gene_count_matrix
  - abundance_matrix
optional_outputs:
  - transcript_count_matrix
output_formats:
  - CSV
  - TSV
  - RDS
depends_on:
  - rnaseq_07_salmon_quantification
can_be_skipped: true
skip_conditions:
  - user chooses alignment-based featureCounts route
default_tool_or_algorithm: tximport
fallback_tool_or_algorithm: tximeta
tool_selection_logic: tximport is simple and stable. tximeta is fallback when transcriptome
                     provenance metadata is needed.
user_settable_parameters:
  - summarization_level
expert_parameters:
  - counts_from_abundance
internal_parameters:
  - tx2gene_source
qc_metrics:
  - genes_detected
  - samples_imported
  - missing_quant_files
qc_pass_fail_rules:
  - fail if any expected sample quantification is missing
  - fail if transcript IDs cannot map to genes
common_failure_modes:
  - transcript ID version mismatch
  - wrong annotation release
  - missing tx2gene mapping
nextflow_module_name: rnaseq/tximport
reusable_for_other_omics: RNA_SEQ_SPECIFIC
```

downstream_steps:

- rnaseq_12_count_matrix_qc_filtering

notes_for_guided_flowchart: Explain this as "make gene count table".

notes_for_dynamic_flowchart: Accepts transcript_quant and emits gene_counts.

notes_for_chat_mode: Chat should include this automatically after Salmon when DESeq2 is requested.

09 STAR Alignment

step_id: rnaseq_09_star_alignment

step_name: Splice-aware genome alignment

category: RNA_SEQ_SPECIFIC

purpose: Align RNA-seq reads to the genome and generate alignment files.

why_this_step_exists: Alignment route is needed for BAM outputs, genome-level QC, splice-aware analysis, and conventional counting.

required_inputs:

- analysis_fastq_manifest
- star_genome_index
- strandedness_call

optional_inputs:

- gtf_annotation

input_formats:

- FASTQ.GZ
- STAR index directory
- GTF

required_outputs:

- aligned_bam_unsorted_or_sorted
- alignment_log

optional_outputs:

- splice_junctions

output_formats:

- BAM
- LOG
- SJ.out.tab

depends_on:

- rnaseq_03_raw_read_qc
- rnaseq_06_strandedness_inference
- rnaseq_06a_reference_bundle_validation

can_be_skipped: true

skip_conditions:

- user chooses Salmon route

default_tool_or_algorithm: STAR

fallback_tool_or_algorithm: HISAT2

tool_selection_logic: STAR is default for standard bulk RNA-seq because it is fast and mature for splice-aware alignment. HISAT2 is fallback for lower memory environments or specific legacy compatibility.

user_settable_parameters:

- alignment_route

expert_parameters:

- read_files_command
- genome_index_overhang
- two_pass_mode

internal_parameters:

- threads
- genome_index_path

qc_metrics:

- uniquely_mapped_percent
- multimapped_percent

```

- unmapped_percent
- chimeric_reads
qc_pass_fail_rules:
- warn if uniquely mapped percent is low
- fail if BAM output is missing
- warn if high multimapping suggests contamination or wrong reference
common_failure_modes:
- wrong genome build
- missing or mismatched GTF
- insufficient memory
nextflow_module_name: rnaseq/star_align
reusable_for_other_omics: RNA_SEQ_SPECIFIC
downstream_steps:
- rnaseq_10_bam_sort_index
notes_for_guided_flowchart: Choose this route when users request BAM or genome alignment.
notes_for_dynamic_flowchart: Emits BAM and alignment metrics.
notes_for_chat_mode: Chat should not choose alignment route unless user needs BAM, splice,
or alignment QC.

```

10 BAM Sort And Index

```

step_id: rnaseq_10_bam_sort_index
step_name: BAM sorting and indexing
category: GENOMICS_COMMON
purpose: Produce coordinate-sorted and indexed BAM files for counting, QC, and
visualization.
why_this_step_exists: Many downstream tools require sorted/indexed BAM files.
required_inputs:
- aligned_bam
optional_inputs: []
input_formats:
- BAM
required_outputs:
- sorted_bam
- bam_index
optional_outputs:
- samtools_stats
output_formats:
- BAM
- BAI
- TXT
depends_on:
- rnaseq_09_star_alignment
can_be_skipped: true
skip_conditions:
- aligner directly emits sorted and indexed BAM accepted by downstream tools
default_tool_or_algorithm: samtools sort/index
fallback_tool_or_algorithm: none
tool_selection_logic: samtools is the standard supported implementation.
user_settable_parameters: []
expert_parameters:
- sort_memory
internal_parameters:
- threads
- temporary_directory
qc_metrics:
- bam_records
- index_created

```

```
qc_pass_fail_rules:
  - fail if sorted BAM or index is missing
common_failure_modes:
  - insufficient disk space
  - insufficient memory for sorting
nextflow_module_name: genomics/samtools_sort_index
reusable_for_other_omics: GENOMICS_COMMON
downstream_steps:
  - rnaseq_11_featurecounts_gene_counting
  - rnaseq_18_multiqc_aggregation
notes_for_guided_flowchart: Hide unless alignment route is expanded.
notes_for_dynamic_flowchart: Required edge between alignment and counting.
notes_for_chat_mode: Chat should include it automatically after STAR.
```

11 Gene Counting With featureCounts

```
step_id: rnaseq_11_featurecounts_gene_counting
step_name: Gene-level read counting
category: RNA_SEQ_SPECIFIC
purpose: Count aligned reads overlapping annotated genes.
why_this_step_exists: Differential expression requires a gene-by-sample count matrix.
required_inputs:
  - sorted_bam
  - gtf_annotation
  - strandedness_call
optional_inputs: []
input_formats:
  - BAM
  - GTF
required_outputs:
  - gene_count_matrix
  - featurecounts_summary
optional_outputs: []
output_formats:
  - TSV
  - TXT
depends_on:
  - rnaseq_10_bam_sort_index
  - rnaseq_06_strandedness_inference
can_be_skipped: true
skip_conditions:
  - user chooses Salmon route
default_tool_or_algorithm: featureCounts
fallback_tool_or_algorithm: none
tool_selection_logic: featureCounts is stable, fast, widely used, and produces MultiQC-
  compatible summaries.
user_settable_parameters:
  - strandedness
expert_parameters:
  - feature_type
  - attribute_type
  - count_multimapping_reads
  - count_overlapping_features
internal_parameters:
  - threads
qc_metrics:
  - assigned_reads_percent
  - unassigned_no_feature_percent
```

```

- unassigned_ambiguity_percent
qc_pass_fail_rules:
- warn if assigned reads percent is low
- fail if count matrix is empty
common_failure_modes:
- wrong strandedness
- wrong annotation release
- chromosome naming mismatch between BAM and GTF
nextflow_module_name: rnaseq/featurecounts
reusable_for_other_omics: RNA_SEQ_SPECIFIC
downstream_steps:
- rnaseq_12_count_matrix_qc_filtering
notes_for_guided_flowchart: Explain as "count reads per gene".
notes_for_dynamic_flowchart: Accepts sorted_bam and emits gene_counts.
notes_for_chat_mode: Chat should include this only for alignment route.

```

12 Count Matrix QC And Filtering

```

step_id: rnaseq_12_count_matrix_qc_filtering
step_name: Count matrix QC and low-expression filtering
category: STATISTICS_COMMON
purpose: Check gene count matrix quality and remove genes with insufficient counts for
reliable testing.
why_this_step_exists: Low-count genes inflate multiple testing burden and add noise.
required_inputs:
- gene_count_matrix
- validated_sample_sheet
optional_inputs: []
input_formats:
- CSV
- TSV
required_outputs:
- filtered_count_matrix
- count_qc_report
optional_outputs:
- normalized_preview_matrix
output_formats:
- CSV
- TSV
- HTML
depends_on:
one_of:
- rnaseq_08_tximport_gene_summarization
- rnaseq_11_featurecounts_gene_counting
can_be_skipped: false
skip_conditions: []
default_tool_or_algorithm: DESeq2 prefiltering logic
fallback_tool_or_algorithm: edgeR filterByExpr
tool_selection_logic: Use DESeq2-compatible filtering by default when DESeq2 is downstream;
use edgeR filterByExpr when edgeR is selected.
user_settable_parameters:
- minimum_count
- minimum_samples
expert_parameters:
- filter_strategy
internal_parameters:
- design_formula
qc_metrics:

```

- genes_before_filtering
- genes_after_filtering
- library_size_per_sample
- zero_count_fraction

qc_pass_fail_rules:

- warn if many samples have very low library size
- warn if most genes are filtered out
- fail if fewer than minimum samples remain

common_failure_modes:

- sample sheet and count matrix sample names do not match
- severe outlier sample
- wrong count source

nextflow_module_name: statistics/count_matrix_qc

reusable_for_other_omics: STATISTICS_COMMON

downstream_steps:

- rnaseq_13_sample_level_qc_pca
- rnaseq_14_differential_expression

notes_for_guided_flowchart: Show how many genes and samples remain.

notes_for_dynamic_flowchart: Emits filtered_gene_counts.

notes_for_chat_mode: Chat should flag sample/count mismatches before DE.

13 Sample-Level QC And PCA

step_id: rnaseq_13_sample_level_qc_pca

step_name: Sample-level QC and PCA

category: STATISTICS_COMMON

purpose: Identify sample outliers, batch effects, and broad structure before differential testing.

why_this_step_exists: Differential expression can be invalid if samples cluster by batch, quality, or hidden covariates.

required_inputs:

- filtered_count_matrix
- validated_sample_sheet

optional_inputs:

- normalized_counts

input_formats:

- CSV
- TSV

required_outputs:

- pca_plot
- sample_distance_plot
- sample_qc_table

optional_outputs:

- interactive_pca_html

output_formats:

- PNG
- PDF
- HTML
- CSV

depends_on:

- rnaseq_12_count_matrix_qc_filtering

can_be_skipped: false

skip_conditions: []

default_tool_or_algorithm: DESeq2 variance-stabilized PCA

fallback_tool_or_algorithm: plotly export for interactive product view

tool_selection_logic: Use DESeq2/R outputs for reproducibility; use Plotly only when interactive UI output is needed.

user_settable_parameters:

```

- color_by
- shape_by
expert_parameters:
- transformation_method
internal_parameters:
- plot_theme
qc_metrics:
- percent_variance_pc1
- percent_variance_pc2
- sample_distance_outliers
qc_pass_fail_rules:
- warn if samples cluster by batch more strongly than condition
- warn if obvious outlier sample is detected
common_failure_modes:
- confounded design
- mislabeled sample
- hidden batch effect
nextflow_module_name: statistics/sample_qc_pca
reusable_for_other_omics: STATISTICS_COMMON
downstream_steps:
- rnaseq_14_differential_expression
notes_for_guided_flowchart: Show this as a review checkpoint.
notes_for_dynamic_flowchart: Can be inserted before most matrix-based statistics.
notes_for_chat_mode: Chat should summarize possible outliers but not remove samples
automatically.

```

14 Differential Expression

```

step_id: rnaseq_14_differential_expression
step_name: Differential expression testing
category: STATISTICS_COMMON
purpose: Identify genes with statistically significant expression differences between
conditions.
why_this_step_exists: This is the primary statistical analysis for many bulk RNA-seq
studies.
required_inputs:
- filtered_count_matrix
- validated_sample_sheet
- contrast_definition
optional_inputs:
- batch_column
input_formats:
- CSV
- TSV
- JSON
required_outputs:
- differential_expression_table
- normalized_counts
- model_summary
optional_outputs:
- shrinkage_results
output_formats:
- CSV
- TSV
- RDS
- JSON
depends_on:
- rnaseq_12_count_matrix_qc_filtering

```

```

- rnaseq_13_sample_level_qc_pca
can_be_skipped: true
skip_conditions:
- user only wants quantification/count matrix
default_tool_or_algorithm: DESeq2
fallback_tool_or_algorithm: edgeR
tool_selection_logic: DESeq2 is default for standard bulk RNA-seq because it is widely
  adopted and has strong shrinkage/modeling support. edgeR is fallback for specific designs
  or edgeR-preferred statistical workflows.
user_settable_parameters:
- condition_column
- reference_condition
- contrast
- batch_column
- fdr_threshold
expert_parameters:
- design_formula
- lfc_shrinkage
- independent_filtering
internal_parameters:
- model_output_prefix
qc_metrics:
- genes_tested
- significant_genes
- dispersion_fit_status
- samples_in_model
qc_pass_fail_rules:
- fail if condition column is missing
- fail if contrast cannot be constructed
- warn if fewer than 3 biological replicates per condition
- warn if design matrix is not full rank
common_failure_modes:
- batch confounded with condition
- too few replicates
- sample names mismatch
- using normalized non-count data as input
nextflow_module_name: statistics/deseq2
reusable_for_other_omics: STATISTICS_COMMON
downstream_steps:
- rnaseq_15_ora_enrichment
- rnaseq_16_gsea_enrichment
- rnaseq_17_visualization_volcano_heatmap
notes_for_guided_flowchart: Ask for comparison in biological language.
notes_for_dynamic_flowchart: Requires count matrix and validated sample metadata.
notes_for_chat_mode: Chat may propose design formula but must ask for confirmation before
  run.

```

15 ORA Enrichment

```

step_id: rnaseq_15_ora_enrichment
step_name: Over-representation enrichment analysis
category: STATISTICS_COMMON
purpose: Test whether significant genes are enriched for pathways or gene sets.
why_this_step_exists: Converts long DE gene lists into interpretable biological themes.
required_inputs:
- differential_expression_table
- gene_set_database
optional_inputs:

```

```

- gene_universe
input_formats:
- CSV
- TSV
- GMT
required_outputs:
- ora_enrichment_table
- ora_plot
optional_outputs:
- leading_gene_list
output_formats:
- CSV
- PNG
- PDF
depends_on:
- rnaseq_14_differential_expression
can_be_skipped: true
skip_conditions:
- user only wants DE table
default_tool_or_algorithm: clusterProfiler ORA
fallback_tool_or_algorithm: g:Profiler API/offline export
tool_selection_logic: Use clusterProfiler by default for local reproducible ORA. Use
  g:Profiler only when the platform supports its API/offline export policy.
user_settable_parameters:
- gene_set_collection
- fdr_threshold
- log2fc_threshold
expert_parameters:
- universe_definition
- min_gene_set_size
- max_gene_set_size
internal_parameters:
- gene_id_mapping_strategy
qc_metrics:
- input_gene_count
- mapped_gene_count
- enriched_term_count
qc_pass_fail_rules:
- warn if many genes fail ID mapping
- warn if significant gene list is too small
common_failure_modes:
- wrong gene identifier namespace
- no explicit universe/background
- outdated pathway database
nextflow_module_name: statistics/clusterprofiler_ora
reusable_for_other_omics: STATISTICS_COMMON
downstream_steps:
- rnaseq_19_report_rendering
notes_for_guided_flowchart: Explain ORA as "pathways overrepresented among significant
  genes".
notes_for_dynamic_flowchart: Separate from GSEA because inputs and assumptions differ.
notes_for_chat_mode: Chat should choose ORA when user asks about significant gene lists.

```

16 GSEA Enrichment

```

step_id: rnaseq_16_gsea_enrichment
step_name: Gene set enrichment analysis
category: STATISTICS_COMMON

```

```
purpose: Test whether ranked genes show coordinated pathway-level shifts.
why_this_step_exists: GSEA can detect pathway signal even when individual genes do not pass
  a hard DE threshold.
required_inputs:
  - ranked_gene_table
  - gene_set_database
optional_inputs: []
input_formats:
  - CSV
  - TSV
  - GMT
required_outputs:
  - gsea_enrichment_table
  - enrichment_plots
optional_outputs:
  - leading_edge_genes
output_formats:
  - CSV
  - PNG
  - PDF
depends_on:
  - rnaseq_14_differential_expression
can_be_skipped: true
skip_conditions:
  - user only wants DE table
default_tool_or_algorithm: fgsea
fallback_tool_or_algorithm: clusterProfiler GSEA
tool_selection_logic: fgsea is default for fast preranked GSEA. clusterProfiler GSEA is
  fallback for teams already using clusterProfiler outputs.
user_settable_parameters:
  - ranking_metric
  - gene_set_collection
  - fdr_threshold
expert_parameters:
  - min_gene_set_size
  - max_gene_set_size
  - permutation_strategy
internal_parameters:
  - gene_id_mapping_strategy
qc_metrics:
  - ranked_genes
  - mapped_ranked_genes
  - significant_gene_sets
qc_pass_fail_rules:
  - warn if too few ranked genes map to gene sets
  - fail if ranking metric cannot be computed
common_failure_modes:
  - duplicate gene IDs
  - wrong ranking direction
  - mismatched species database
nextflow_module_name: statistics/fgsea
reusable_for_other_omics: STATISTICS_COMMON
downstream_steps:
  - rnaseq_19_report_rendering
notes_for_guided_flowchart: Explain as "pathway shifts across all ranked genes".
notes_for_dynamic_flowchart: Do not merge with ORA; keep as separate analytical route.
notes_for_chat_mode: Chat should choose GSEA when user asks about pathway-level trends.
```

17 Visualization

```
step_id: rnaseq_17_visualization_volcano_heatmap
step_name: RNA-seq result visualization bundle
category: REPORTING_COMMON
purpose: Generate standard figures for interpreting differential expression results.
why_this_step_exists: Users need interpretable plots, but each plot should be task-specific
                    rather than a generic plotting toolbox.
required_inputs:
  - differential_expression_table
  - normalized_counts
optional_inputs:
  - gene_annotations
input_formats:
  - CSV
  - TSV
required_outputs:
  - volcano_plot
  - ma_plot
  - heatmap_plot
optional_outputs:
  - top_gene_table
output_formats:
  - PNG
  - PDF
  - HTML
  - CSV
depends_on:
  - rnaseq_14_differential_expression
can_be_skipped: true
skip_conditions:
  - user only wants machine-readable tables
default_tool_or_algorithm: task-specific R plotting functions for volcano, MA, and heatmap
                           outputs
fallback_tool_or_algorithm: Plotly for interactive UI exports
tool_selection_logic: Use standard R plotting for reproducible static reports; use Plotly
                    only for interactive application views.
user_settable_parameters:
  - fdr_threshold
  - log2fc_threshold
  - top_n_genes
expert_parameters:
  - gene_label_strategy
  - heatmap_scaling_method
internal_parameters:
  - plot_theme
  - color_palette
qc_metrics:
  - plots_generated
  - genes_available_for_heatmap
qc_pass_fail_rules:
  - warn if too few significant genes for heatmap
  - fail if required DE table columns are missing
common_failure_modes:
  - missing adjusted p-value column
  - gene symbol mapping failure
  - too many labels causing unreadable plots
nextflow_module_name: reporting/rnaseq_visualization_bundle
reusable_for_other_omics: REPORTING_COMMON
```

downstream_steps:

- rnaseq_19_report_rendering

notes_for_guided_flowchart: Display figures as interpretation outputs, not analysis decisions.

notes_for_dynamic_flowchart: Production UI should expose volcano, MA plot, and heatmap as child nodes even if the backend runs one reporting bundle.

notes_for_chat_mode: Chat can propose figures but should not alter thresholds without confirmation.

18 MultiQC Aggregation

step_id: rnaseq_18_multiqc_aggregation

step_name: MultiQC aggregation

category: REPORTING_COMMON

purpose: Aggregate QC logs and reports into one run-level QC report.

why_this_step_exists: Production users need one place to review QC across all samples and tools.

required_inputs:

- fastqc_outputs

optional_inputs:

- trimming_reports
- alignment_logs
- quantification_logs
- featurecounts_summary
- custom_multiqc_config

input_formats:

- HTML
- ZIP
- TXT
- JSON

required_outputs:

- multiqc_html
- multiqc_data

optional_outputs:

- multiqc_plots

output_formats:

- HTML
- TXT
- JSON
- PNG

depends_on:

- rnaseq_03_raw_read_qc

can_be_skipped: false

skip_conditions: []

default_tool_or_algorithm: MultiQC

fallback_tool_or_algorithm: none

tool_selection_logic: MultiQC is the supported QC aggregation standard.

user_settable_parameters:

- report_title

expert_parameters:

- custom_multiqc_config

internal_parameters:

- search_directory
- module_order

qc_metrics:

- modules_detected
- samples_in_report
- missing_qc_sections

```
qc_pass_fail_rules:
  - warn if expected modules are missing
  - fail if no QC files are detected
common_failure_modes:
  - tool output paths not published
  - unsupported output format
  - sample naming mismatch across tools
nextflow_module_name: common/multiqc
reusable_for_other_omics: REPORTING_COMMON
downstream_steps:
  - rnaseq_19_report_rendering
notes_for_guided_flowchart: Show as "combined QC report".
notes_for_dynamic_flowchart: Aggregates any compatible QC artifacts.
notes_for_chat_mode: Chat should summarize MultiQC warnings in plain language.
```

19 Report Rendering

```
step_id: rnaseq_19_report_rendering
step_name: Final report rendering
category: REPORTING_COMMON
purpose: Produce a human-readable analysis report with methods, QC, figures, and result
  summaries.
why_this_step_exists: A production workflow needs a reproducible report, not only raw output
  files.
required_inputs:
  - multiqc_html
  - run_manifest
  - tool_versions
optional_inputs:
  - differential_expression_table
  - visualizations
  - enrichment_results
  - user_notes
input_formats:
  - HTML
  - CSV
  - TSV
  - JSON
  - PNG
  - PDF
required_outputs:
  - final_report
optional_outputs:
  - methods_section
  - downloadable_summary
output_formats:
  - HTML
  - PDF
  - DOCX
depends_on:
  - rnaseq_18_multiqc_aggregation
can_be_skipped: true
skip_conditions:
  - machine-readable run only
default_tool_or_algorithm: Quarto or R Markdown
fallback_tool_or_algorithm: none
tool_selection_logic: Use one reproducible reporting system to avoid template drift.
user_settable_parameters:
```

```
- report_title
- include_methods
- include_qc
- include_enrichment
expert_parameters:
- report_template
internal_parameters:
- artifact_paths
- render_profile
qc_metrics:
- report_sections_rendered
- missing_artifacts
qc_pass_fail_rules:
- fail if required report artifacts are missing
- warn if optional sections cannot render
common_failure_modes:
- missing figure paths
- invalid table schema
- rendering dependency failure
nextflow_module_name: reporting/render_report
reusable_for_other_omics: REPORTING_COMMON
downstream_steps:
- rnaseq_20_provenance_export
notes_for_guided_flowchart: Final user-facing deliverable.
notes_for_dynamic_flowchart: Can accept outputs from many assay workflows.
notes_for_chat_mode: Chat may draft interpretation but report generation requires user
confirmation.
```

20 Provenance And Artifact Export

```
step_id: rnaseq_20_provenance_export
step_name: Provenance and artifact export
category: COMMON
purpose: Export final artifacts, run metadata, tool versions, parameters, and audit-ready
provenance.
why_this_step_exists: Results must be reproducible, inspectable, and traceable.
required_inputs:
- run_manifest
- nextflow_trace
- tool_versions
- final_outputs
optional_inputs:
- audit_context
input_formats:
- JSON
- TXT
- CSV
- HTML
required_outputs:
- artifact_manifest
- provenance_manifest
- published_results_directory
optional_outputs:
- ro_crate
output_formats:
- JSON
- directory
- RO-Crate
```

```

depends_on:
  - rnaseq_19_report_rendering
can_be_skipped: false
skip_conditions: []
default_tool_or_algorithm: custom Survom run manifest exporter
fallback_tool_or_algorithm: R0-Crate export
tool_selection_logic: Use platform manifest for product needs; offer R0-Crate when
  interoperability is required.
user_settable_parameters:
  - export_format
expert_parameters:
  - include_intermediate_files
internal_parameters:
  - object_storage_prefix
  - audit_actor
  - retention_policy
qc_metrics:
  - artifacts_published
  - manifest_complete
  - tool_versions_recorded
qc_pass_fail_rules:
  - fail if final required outputs are not published
  - fail if tool versions are missing
common_failure_modes:
  - incomplete run due to failed upstream process
  - object storage permission error
  - missing software version capture
nextflow_module_name: common/provenance_export
reusable_for_other_omics: COMMON
downstream_steps: []
notes_for_guided_flowchart: Usually hidden; shown as "results saved".
notes_for_dynamic_flowchart: Terminal node for all workflows.
notes_for_chat_mode: Chat can explain where outputs are stored and what parameters were
  used.

```

Senior Review Notes For RNA-seq Tutorial Materials

Bottom Line

Do not share design/rna-seq/first_try.md as the final student-facing or reviewer-facing document.

It is useful as a source draft, but it reads like a tool encyclopedia. A senior reviewer may question it because it exposes too many tools, mixes production workflow design with low-level command flags, and includes several claims that should be verified against official tool documentation before publication.

Use these two curated files instead:

- docs/rnaseq_atomic_step_registry_curated.md
- docs/rnaseq_tool_selection_policy.md

Main Problems In The Draft

1. It starts directly with raw read QC and does not clearly define project setup, sample-sheet validation, run metadata, or reference validation as first-class production steps.
2. It lists too many tools per step. That is useful for teaching tool history, but weak for a production platform. The platform should expose one default and one fallback per step.
3. Reference compatibility is not emphasized enough. Genome FASTA, GTF, STAR index, Salmon index, transcriptome FASTA, and tx2gene mapping must be checked together. This is one of the easiest places to produce silently wrong RNA-seq results.
4. Some tool claims are too specific for an architecture document, for example exact speed claims, default flags, and precise option behavior. These should be marked `VERIFY_IN_OFFICIAL_DOCS` unless checked.
5. Trimming includes methylation/RRBS-specific guidance. That belongs in a methylation workflow note, not in the main RNA-seq MVP path.
6. The document mixes alignment-based RNA-seq, pseudoalignment RNA-seq, reference building, QC, DE, enrichment, and visualization without a strong route model. Students may copy the wrong combination of steps.
7. Visualization is treated as one broad plotting step. For a dynamic workflow UI, volcano, MA plot, heatmap, PCA, and enrichment plots should be separate visible tasks or child nodes.
8. MultiQC inputs must be route-aware. A Salmon-only run should not require STAR or featureCounts logs.
9. Report generation should not depend on provenance output if provenance is produced after the report. That creates a circular workflow dependency.
10. Differential expression guidance should repeatedly state that DESeq2/edgeR require raw integer counts, not TPM, FPKM, RPKM, or normalized values.

Corrections Already Applied

The curated registry now includes:

- explicit reference-bundle validation,
- route-aware Salmon and STAR dependencies,
- `analysis_fastq_manifest` instead of ambiguous raw/trimmed FASTQ input,
- one-of dependency for count matrix QC from either `tximport` or `featureCounts`,
- route-aware MultiQC optional inputs,
- report rendering without a circular provenance dependency,
- clearer visualization-bundle wording for production UI design.

What Is Safe To Give Students

Give students:

- the curated atomic registry,
- the tool selection policy,
- a small route diagram,
- one clean hands-on exercise using Salmon -> `tximport` -> DESeq2,
- one optional alignment route exercise using STAR -> `featureCounts` -> DESeq2.

Keep `first_try.md` as an internal reference draft only.